

# Railtrack Company Code of Practice

This document is the property of Railtrack PLC, part of the Network Rail group of companies.

It shall not be reproduced in whole or in part, nor disclosed to a third party, without the written permission of the Standard Owner.

© Copyright 2003 Railtrack PLC, part of the Network Rail group of companies.

## METHODOLOGY FOR THE DETERMINATION OF INTERACTION WITH NEIGHBOURING RAILWAYS

### ENDORSEMENT & AUTHORISATION

Endorsed by:



A Simmons, Professional Head of Signal Engineering

Accepted for issue by:



P McAleer, Acting Head of Business Standards

Authorised by:



A McNaughton, Chief Engineer

**ISSUE RECORD**

| <b>ISSUE</b> | <b>DATE</b>          | <b>COMMENTS</b>             |
|--------------|----------------------|-----------------------------|
| <b>I</b>     | <b>February 2003</b> | <b>New Code of Practice</b> |

**IMPLEMENTATION**

The provisions of this code of practice may be used from the date of issue.

**DISCLAIMER**

Railtrack PLC, part of the Network Rail group of companies, has used its best endeavours to ensure that the content, layout and text of this document are accurate, complete and suitable for its stated purpose. It makes no warranties, express or implied, that compliance with the contents of this document shall be sufficient to ensure safe systems of work or operation. Railtrack PLC, part of the Network Rail group of companies, will not be liable to pay compensation in respect of the content or subsequent use of this document for any purpose other than its stated purpose or for any purpose other than that for which it was prepared except where it can be shown to have acted in bad faith or there has been wilful default.

**SUPPLY**

Copies of documents are available electronically, within Railtrack's organisation. Hard copies of this document will be available to Railtrack staff on request to the Document Controller and to other organisations from Technical Indexes Ltd (01344 404409).

Published & Issued by **Railtrack PLC (part of the Network Rail group of companies)**  
**Railtrack House**  
**Euston Square**  
**LONDON**  
**NW1 2EE**

## Contents

|                                                   |           |
|---------------------------------------------------|-----------|
| <b>1. OBJECTIVES .....</b>                        | <b>4</b>  |
| 1.1 PURPOSE .....                                 | 4         |
| 1.2 SCOPE .....                                   | 4         |
| <b>2. SOURCE DOCUMENTS.....</b>                   | <b>5</b>  |
| 2.1 GROUP AND LINE STANDARDS .....                | 5         |
| 2.2 PREVIOUS SAFETY CASE DOCUMENTATION .....      | 5         |
| 2.3 OTHER STANDARDS .....                         | 6         |
| 2.4 OTHER DOCUMENTATION .....                     | 6         |
| <b>3. INFRASTRUCTURE ASSUMPTIONS .....</b>        | <b>6</b>  |
| <b>4. METHODOLOGY .....</b>                       | <b>7</b>  |
| 4.1 IDENTIFY NEIGHBOURING SYSTEMS.....            | 7         |
| 4.2 DETAILS REQUIRED .....                        | 8         |
| 4.3 CHARACTERISATION .....                        | 8         |
| 4.4 DETERMINE THE COUPLING FACTORS .....          | 9         |
| 4.4.1 <i>Calculations</i> .....                   | 9         |
| 4.4.2 <i>Practical Tests</i> .....                | 10        |
| 4.5 DEMONSTRATION OF COMPLIANCE .....             | 10        |
| 4.6 SUBMISSION .....                              | 11        |
| <b>APPENDIX A: MODELLING .....</b>                | <b>12</b> |
| A1 TYPES OF INTERFACES .....                      | 12        |
| A2 EQUIVALENT CIRCUIT FOR GENERAL INTERFACE ..... | 12        |
| A3 SHARED ELECTRICAL INTERFACE .....              | 22        |
| A4 WORKED EXAMPLE.....                            | 22        |
| <b>APPENDIX B: ASSUMPTIONS .....</b>              | <b>24</b> |
| <b>APPENDIX C: SOURCES OF TECHNICAL WORK.....</b> | <b>25</b> |

## **I. OBJECTIVES**

### **I.1 Purpose**

The purpose of this code of practice is to provide a methodology to demonstrate compatibility of traction and rolling stock operating on Railtrack electrified lines with the infrastructure of neighbouring railways & vice versa. This document is based on several submissions to railways the neighbour Railtrack lines and it should not be inferred that it has been approved by any particular neighbour neither does its use guarantee acceptance by any neighbour. The authority that constitutes the 'neighbouring' party should be consulted and approval to use the methodology presented here gained in principle.

### **I.2 Scope**

This document is a generic EMC approach to be adopted for assessing compatibility between trains on Railtrack and systems on neighbouring railways' infrastructure.

Demonstration of compliance is achieved by the following process:

- Identify neighbouring systems.
- Determine system susceptibility.
- Establish coupling factors.
- Determine allowable levels on Railtrack.
- Assess permissible levels against train emissions.

These elements allow the wheel/rail interface currents to be defined on the Railtrack infrastructure. These can then be compared to the train emissions under defined operational conditions at the input terminals of the equipment on neighbouring infrastructure after taking into account the possible coupling factors between the adjacent railways. Thus the overall approach for demonstrating compatibility provided in the report is based on level alone.

The appendix to this standard which relates to modelling considers only conductive and inductive coupling. These coupling mechanisms are generally dominant, this should be confirmed for the system under study.

The interference level due to energy coupling to a neighbouring railway's equipment is assessed and compared against existing permissible levels. This assessment includes interference, which has the potential to affect track circuits and other equipment

installed on the neighbouring railway, generated by trains running on adjacent Railtrack lines, when operating normally and when they experience failures. The possibility of Radio Frequency interference shall also be considered, guidance being given in the ITU Directives, ref. [15].

## **2. SOURCE DOCUMENTS**

### **2.1 Group and Line Standards**

1. Electrified Lines Traction Bonding, GL/RT1254, Issue I, April 2000.
2. Railtrack Line Code of Practice - Design and Installation of Negative Bonding and Associated Equipment on D.C. Electrified Lines, RT/E/C/27022, Issue I, March 1998.
3. Railtrack Line Specification - Design of Earthing and Bonding Systems for 25 kV A.C. Electrified Lines, RT/E/S/21085, Issue I, October 1998.

### **2.2 Previous Safety Case documentation**

4. 'Class 92 Locomotive - EMC Interface between Class 92 A.C. Routes and London Underground Ltd', AEAT Rail Ref. ESI9009, Issue 20, November 1999.
5. 'Class 92 Locomotive - EMC Interface between Class 92 A.C. Routes and Docklands Light Railway', AEAT Rail Ref. ESI9025, Issue 20, November 1999.
6. 'Class 92 Locomotive EMC Safety Assessment, Fleet Operation on the 25KV AC Railway between Crewe and WEFOC, Vol.4 - LUL Boundaries', AEAT Rail Ref. DR0483, Issue A0, June 1999.
7. 'LUL Interfaces report for Class 375 "Electrostar" Operating on Connex South Eastern Routes', AEAT Rail Ref. LD00421-I, Issue 1a, February 2000.
8. 'Electric Traction Interference: London Underground and Docklands Light Railway, Juniper Project', Alstom Ref. 10109.4112/Rp17, Issue 4, October 1998.
9. 'Class 357 EMU Compatibility with LUL Infrastructure', Adtranz Ref. 3EER300000-5305, Revision \_C 2000-05-03, February 2001.

10. "Class 92 Fleet Operation on the 25 kV AC Railway between Crewe and Doncaster – Reed FDM", AEA Technology Rail ref. I20-dr0458, version b0, dated November 1999.

## 2.3 Other Standards

11. prEN 50238 'Railway Applications: Compatibility between Rolling Stock and Train Detection Systems' Final draft 1998.
12. BS: EN 50121 'Railway Applications: Electromagnetic Compatibility', 2000.

## 2.4 Other Documentation

13. Engineering Safety Management, Yellow Book 3, Issue 3, Railtrack PLC 2000
14. S. Ramo, Whinnery J.R., Duzer T.V., Fields and Waves in Communication Electronics, 3<sup>rd</sup> ed., John Wiley & Sons, Inc., USA, 1994.
15. Directives concerning the protection of telecommunication lines against harmful effects from electric power and electrified railway lines, Vol.II - Calculating induced voltages and currents in practical cases, ITU-T, 1999.
16. Erling D Sunde, "Earth Conduction Effects in Transmission Systems", Bell Telephone Laboratories, 1968.
17. Kraus J.D., "Electromagnetics", McGraw - Hill, Fourth edition, 1996.
18. Tagg G.F, "Earth Resistances", George Newnes Limited, London, 1964.
19. J.C.Brown, J. Allan, B.Mellitt, 'Calculations and Measurements of Rail Impedances Applicable to Remote Short Circuit Fault Currents', IEE Proceedings, July 1992.

## 3. INFRASTRUCTURE ASSUMPTIONS

The infrastructure assumed during the analysis is as described below, further assumptions are listed in Appendix B:

- The equipment is operating within limits specified by the manufacturer, the design authority or by the infrastructure controller.

- The equipment is correctly installed in accordance with applicable handbook of the time and mandatory subsequent amendments.
- The equipment is correctly maintained in accordance with the relevant standards and procedures.
- The exposure distance between the two railways is equal to the maximum length of the third party's track circuit considered. This assumption is only used for the 'general interface' case.
- The traction bonding arrangements on the RT lines are compliant with [1], [2] and [3].
- The running rails are assumed to be type BSI 13A, normal grade. Check rails are excluded. The sleepers are conventional concrete or wooden type. The track is assumed to be of standard gauge.
- Coupling factors for steady state, transient and fault levels are assumed to be identical.

## 4. Methodology

### 4.1 Identify neighbouring systems

When considering the permissible emission levels for trains on Railtrack infrastructure neighbouring railway systems need to be considered taking into account Railtrack's electrification boundaries, principally these systems are underground, light rail and metro railways. These systems whilst not owned by Railtrack must be assessed and the owners of the system satisfied that their integrity is maintained. The Railtrack approvals process requires that sufficient evidence shall be provided to the owners of the neighbouring systems to enable them to do this and to ultimately confirm their acceptance to the proposed change on Railtrack's lines.

The boundary to be considered for interference in a route safety case is defined by line of route and off route boundaries. All neighbouring railways and associated equipment within this boundary form the scope of the assessment. Neighbouring railways can either be:

- Physically connected;
- Running in parallel;
- Crossing on or over lines within 1 mile distance.

Details shall be obtained of the systems installed on the neighbouring infrastructure and written confirmation obtained from the operators to verify the installation details of all relevant interfaces between the two rail infrastructures. Information on any available interference specifications and assumed interference allowances should be sought.

Clearance for running shall be granted by the neighbouring railway party based on a summary document of the background to the project, emission levels (summary of the EMI footprint of the trains) etc. Based on this information the neighbouring administration may need to do their own assessment of acceptability to operate new systems on Railtrack lines. In this case, they usually carry out monitoring of their own circuits during initial operations, to further satisfy themselves that there are no problems relating to operation of the new equipment.

## 4.2 Details required

The EMC assessments can only be conducted if manufacturers' equipment is assumed to have the characteristics declared and published by them and to have been correctly installed and maintained in accordance with existing standards and procedures. In majority of cases compliance with the requirements laid down in [11] and [12] is sufficient. It must also be assumed that the signalling equipment is operating within its required specification and operating limits.

If additional proof is required by the authorities of adjacent railways then the exact nature of the installation and interfaces shall be studied and assessed correspondingly.

The assessments shall consider the effect on the neighbouring railways' systems of EMI resulting from the operation of the new train. The interference mechanisms considered will be conductive and inductive coupling, this includes induced voltages and Radio Frequency Interference (RFI).

In those cases when the neighbouring authority conducts its own assessment a report shall be produced giving the background to the system. Also a summary of the EMI footprint based on testing and comment on the effect of credible failures on the footprint, with specific reference to frequencies of interest for the third party railway shall be prepared and submitted.

## 4.3 Characterisation

Once the neighbouring railways' systems have been identified the equipment that may be susceptible to interference will need to be identified [8].



It is recommended that the assessment of the equipment itself shall be based on the same three main methods defined for the Railtrack EE&CS equipment as follows:

- (i) Characterising the susceptibility of the infrastructure to EMI so as to define the interference requirements for the train at the wheel/rail interface.
- (ii) Compliance with relevant standards.
- (iii) Hazard and Operability (HAZOP) meetings.

The major items of equipment such as track circuits and line side systems on the neighbouring railway need to be assessed in detail and this will be done in accordance with items (i) and (ii) above. The susceptibility of the equipment to interference will be defined in an interface document that will detail the interference limits at the wheel rail interface on Railtrack infrastructure. The interface document will contain a description of the EE&CS equipment, its operating characteristics and the derivation of the interference limits that result from this.

A HAZOP approach may be appropriate for the miscellaneous, telecommunications and electrification equipment. The results of the HAZOP studies should be formally recorded.

A list of reference documents shall be maintained throughout the assessment, principally being the interface specification report, the applicable European standards and Yellow Book 3 [13].

#### **4.4 Determine the coupling factors**

Where further analysis to [(i) - (iii)] described in section 4.3 is considered necessary, the following procedural steps can be followed:

##### **4.4.1 Calculations**

1. Determine the susceptibility of those systems and equipment installed on the neighbouring railways and not defined in 4.3 in terms of susceptible frequencies and operating voltages.
2. Determine the inductive coupling factors between Railtrack's and neighbouring railway's circuits at each of the relevant frequencies identified in step 1 above. This is carried out using Carson - Pollaczek calculations, see

## APPENDIX A: Modelling.

3. Taking into account any allowances and interference levels determine what level of current is permissible on Railtrack infrastructure based on the operating voltages identified above.

An exercise shall be carried out to determine whether or not there are any direct metal work connections between the boundary systems and the adjacent routes. Following this the coupling factors between those routes and the boundary system(s) shall be calculated, this will enable the proportion of any relevant interference produced by the new train that may be coupled into the adjacent neighbouring system to be calculated.

In terms of inductive coupling it has previously been demonstrated that the Carson - Pollaczek formulae for the calculation of mutual coupling are suitable for use within a railway environment [10].

### 4.4.2 *Practical Tests*

Alternatively to 4.4.1, practical tests may be undertaken to demonstrate an acceptable safety margin with respect to the new train operating close to the neighbouring railway. The justification of the margin is based on comparative argument that the EMI produced as a result of operating the new train do not exceed the levels already accepted by authorities during operation of existing rolling stock and as such do not present any additional risk to the neighbouring infrastructure [9].

## 4.5 **Demonstration of Compliance**

The evidence demonstrating the EMC of the new train with the neighbouring railway's infrastructure will be supplied in a set of controlled documents to show that the train under assessment will present only a tolerable or as low as reasonably practicable (ALARP) risk to the neighbouring railways' equipment and circuits. This will achieve the following:

- Define what train operations and corresponding certification is sought.
- Set out what is required to support this.
- Demonstrate that this has been achieved if necessary by referencing and summarising the assessments conducted in the other documents.

In addition to the main submission documents there will be a significant amount of supporting documentation that will contain the technical details relating to the analysis of the train electrical systems and interference profile.

This report will document the calculations and simulations performed and the results obtained. It will also provide an audit trail for the theoretical analysis.

#### **4.6 Submission**

The aim of the assessment is to gain clearance from the authorities of the neighbouring railways. The review procedure of the neighbouring railway will differ from one authority to another. In general terms the level of clearance for operations should be stated by the neighbouring railway along with any limitations. In the case of operations over Railtrack infrastructure and the neighbour being someone other than Railtrack, the Railtrack Acceptance Body (see [11]) need to be assured that the neighbouring railway has no objections to the operating level being sought.

The above approach ensures that the risk to the neighbouring railways' systems resulting from operation of the new train is negligible or ALARP.

## APPENDIX A: Modelling

The analysis method for right side failure is the same as for wrong side failure. The tolerated interference current used in the analysis should envelop both Right Side Failure (RSF) and Wrong Side Failure (WSF) conditions. Interference budgets shall be established for particular cases. The accuracy of modelling is considered acceptable for both these cases [7].

### A1 Types of Interfaces

The analysis of interference currents is based on two cases of interfaces depending on whether or not there are any metal work connections between the boundary systems and the adjacent routes:

1. General Interface - the rails of the two railways are isolated from each other.
2. Shared electrical interface - common-bonding arrangements need to be considered precisely before commencing the assessment.

### A2 Equivalent Circuit for General Interface

The approach used in the analysis is based on the techniques used in [4]. The methodology provided calculates currents in the running rails of neighbouring railways and as such remains the same for any type of electrification used by them. Specific attention is required to solve the equivalent circuit. The worst case realistic inducing circuit should be assumed. In the majority of cases the worst case is when the 'go' and 'return' conductors of the inducing circuit are the furthest apart. In most cases booster transformers and/or return conductors provide additional path for the earth return current and hence are not considered to be the representative worst case of rail return only. Only conductive and inductive coupling have been considered in the modelling. The capacitive coupling is generally very small and therefore ignored.

The model uses a circuit analysis technique. A schematic diagram showing different configurations between the two railways is shown in Figure A-1. Figure A-2 shows the equivalent circuit diagram used in the analysis. Although Figure A-1 shows 25 kV overhead electrification the methodology is equally applicable to third rail. Fences installed between the two railways should generally be ignored. However, if they are thought to be significant these elements can be treated as an additional conductor running in parallel to the two tracks. A metal fence can be treated as a current carrying conductor having a distributed earth and also mutually coupled to the rest of conductors in the system. For tunnels

constructed using cast iron or steel they can be treated similarly as an additional current carrying conductor which is mutually coupled to the rest of conductors in the system.

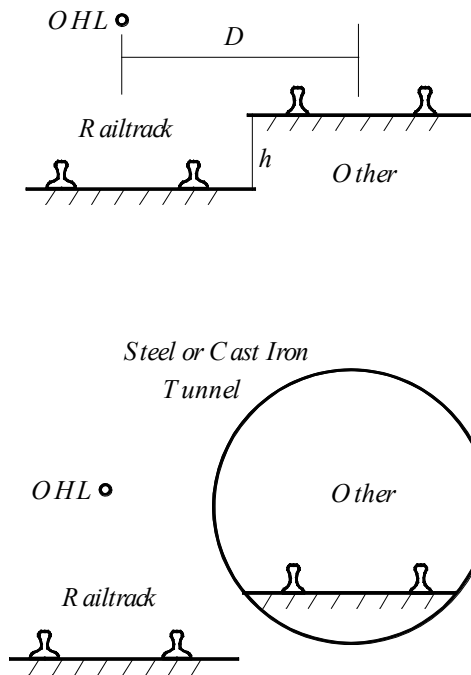
The assessment should be based on determining the worst case of coupling factors between the current flow on the RT infrastructure and current flow in the running rails of the adjacent line. The worst case is for the stretch they run in closest parallel. The worst case will be a track circuit installed at the maximum permitted length exposed for the entirety of its length.

The electrical circuit model is configured with the track split into sections joined together. This allows the effects of mutual coupling and conductive coupling to be distributed along the length of the track as an approximation to the real effects.

The conductive coupling between the tracks is represented by connecting an impedance between (i) the two running rails of each track to represent the conduction through the ballast (ii) an earth impedance between each rail and a common earth point and (iii) an impedance connecting the inner rails of the two tracks to represent the surface conduction. These impedances are connected at the junction of each track section and at both ends of the tracks. The surface resistance and the rail-to-earth resistance will vary according to the earth resistivity and the distance between the tracks. For fences and tunnels the earth impedances are treated in the same way as shown in Figure A-1. This conductive coupling can be modified to take into account a shared electrical interface.

When crossing at an oblique angle, either over a bridge or in a tunnel, both the conductive and inductive coupling for the different sections will change.

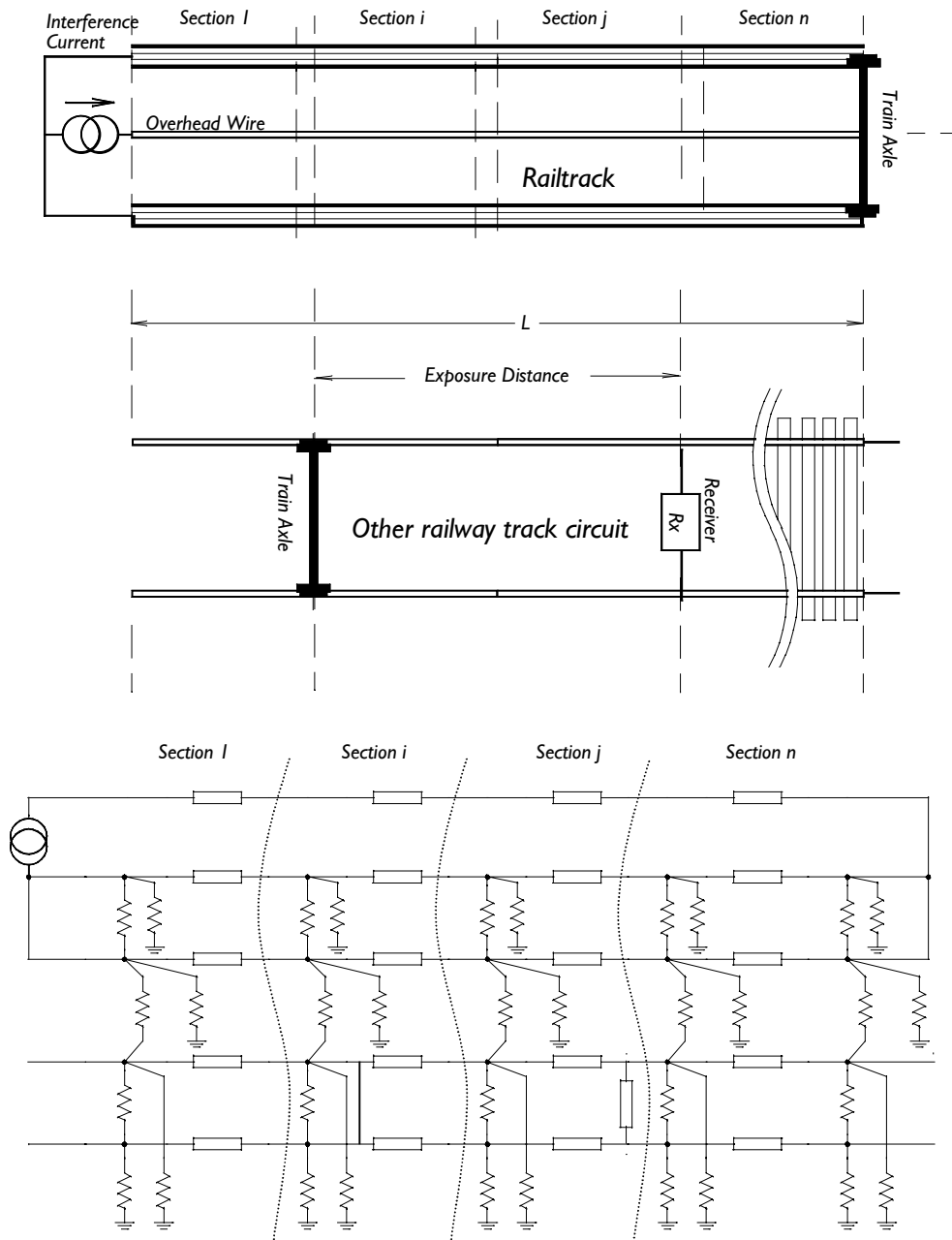
The output from the model is the current flowing in the running rails of the neighbouring railways calculated as the average of a number of sections for the two running rails. The coupling factor is linear and so can be expressed as a percentage or the current that flows in the 'victim' circuit per amp of current in the inducing circuit.



**Figure A-1 Different configurations of Railtrack-other railways interfaces**

The methodology provided calculates currents in the rails of neighbouring railways and as such remains the same for any type of electrification used by them. The rail can be modelled as an equivalent cylindrical conductor, with circumference equal to the rail periphery, for calculating resistance above approximately 12 Hz. Assuming that the boundary between the internal and external rail flux linkages is defined by a circle with the same periphery than the inductance due to flux linkage external to the rail can also be calculated [19].

The interference current on RT lines is assumed to be a current source injected at a certain point and the actual value at each point of the line is calculated by solving the circuit in fig. A-2:



Equivalent Circuit, only 4 sections are shown

Figure A-2 Equivalent circuit diagram at Railtrack-to-other interface

## A2.1 Self Impedance and Inductive Coupling

### Series Impedance of Conductors

The series impedance of the running rails, conductor rails, over head wire and round conductors in the system is determined using the approach given in [4] and summarised as follows:

The impedance of a round steel conductor is defined by equation 1 [14]:

$$Z_i = \frac{(1+j)R_s}{2\pi r_0}, \Omega/m \text{ where:} \quad (1)$$

$R_s = (\sigma\delta)^{-1}$  surface resistivity in  $\Omega$ ;

$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}$  skin depth in m;

$r_0$  conductor outer radius in metres

$\sigma$  conductivity of the conductor in  $1/(\Omega.m)$

$\mu$  permeability in H/m

$\omega$  radians per second,  $2\pi f$

The resistance and internal reactance are equal for ferromagnetic conductor.

For any other type conductor, equation 2 applies for low frequencies:

$$Z_i \approx \frac{1}{\pi r_0^2 \sigma} \left[ 1 + \frac{1}{48} \left( \frac{r_0}{\delta} \right)^2 \right] + j \frac{\omega\mu}{8\pi} \quad (2)$$

The current approach used successfully for Class 92 and 375 safety cases [4] and [7] is based



on the electromagnetic theory and basic equations. The considerations in the reference source are confined to the self- and mutual resistances of cylindrical conductors surrounded by a conducting medium of infinite extend. Longitudinal current is assumed constant along the length. The equation for half buried conductor has been verified against the generally used Laplace's equation where the rail to rail resistance is proportional to the logarithm of the square of the separation distance assuming the earth is a homogenous medium, [17]. The data used in the other IDI documents and which are mostly based on site measurements can also be used.

### Inductive Coupling

The mutual coupling for multiple conductors running parallel with an earth return is determined using the Carson-Pollaczek methodology, [4]. A simplified formula is shown in equation 3:

$$M_{ij} = \frac{\mu_o}{2\pi} \left[ \ln(D_e/D_{ij}) - j\alpha \frac{\pi}{4} \right] \quad (3)$$

where,  $D_e = 658 \sqrt{\frac{\rho}{f}}$  the distance in metres of the effective return earth current,

$M_{ij}$  mutual coupling between conductor  $i$  and  $j$  per metre

$D_{ij}$  distance between conductor  $i$  and  $j$  in metres, refer to Figure B-1

$f$  frequency

$\rho$  earth resistivity in  $\Omega.m$

$\alpha$  proportion of current flowing in the earth ( $< 1$ )

$\mu_o$  absolute permeability  $4\pi * 10^{-7}$  H/m

The resistive coupling is represented by the imaginary part in the brackets of formulae 3. The methodology for using Carlson-Pollaczek equations suggests that iterations should take place when solving the equation so that this part is diminished. The conditions when this can happen correspond to 30% earth return current. This is a very pessimistic assumption.

Further guidance can be found in [15].

## A2.2 Conductive Coupling

The earth conductive coupling is modelled as a pure resistance linking the rails to a common ground point and also linking the different rails in the form of a surface resistance. (Capacitive coupling, if significant may also be introduced by the properties of the earth.) The bonding to earthing points are modelled as a resistance to the common ground point. In modelling these components the earth is represented by a homogenous media having a constant earth resistivity,  $\rho$ , given in  $\Omega.m$ . The different combinations of earth resistance required can be classified as follows, refer to Figure A-3:

- Resistance of a floating single rail to the earth point per unit length
- Resistance of multiple floating rails to the earth point per unit length
- Resistance of bonding and earthing points to the earth point
- Resistance of a single rail earthed at regular intervals to the earth point per unit length
- Resistance of multiple rails earthed at regular intervals to the earth point per unit length
- Resistance of the rail to rail earth for different combinations of single and/or multiple rails, either floating or earthed, per unit length
- Resistance of a fence to the earth point per unit length
- Resistance of a tunnel to the earth point per unit length

The equations are taken from [16] and considered acceptable in [5].

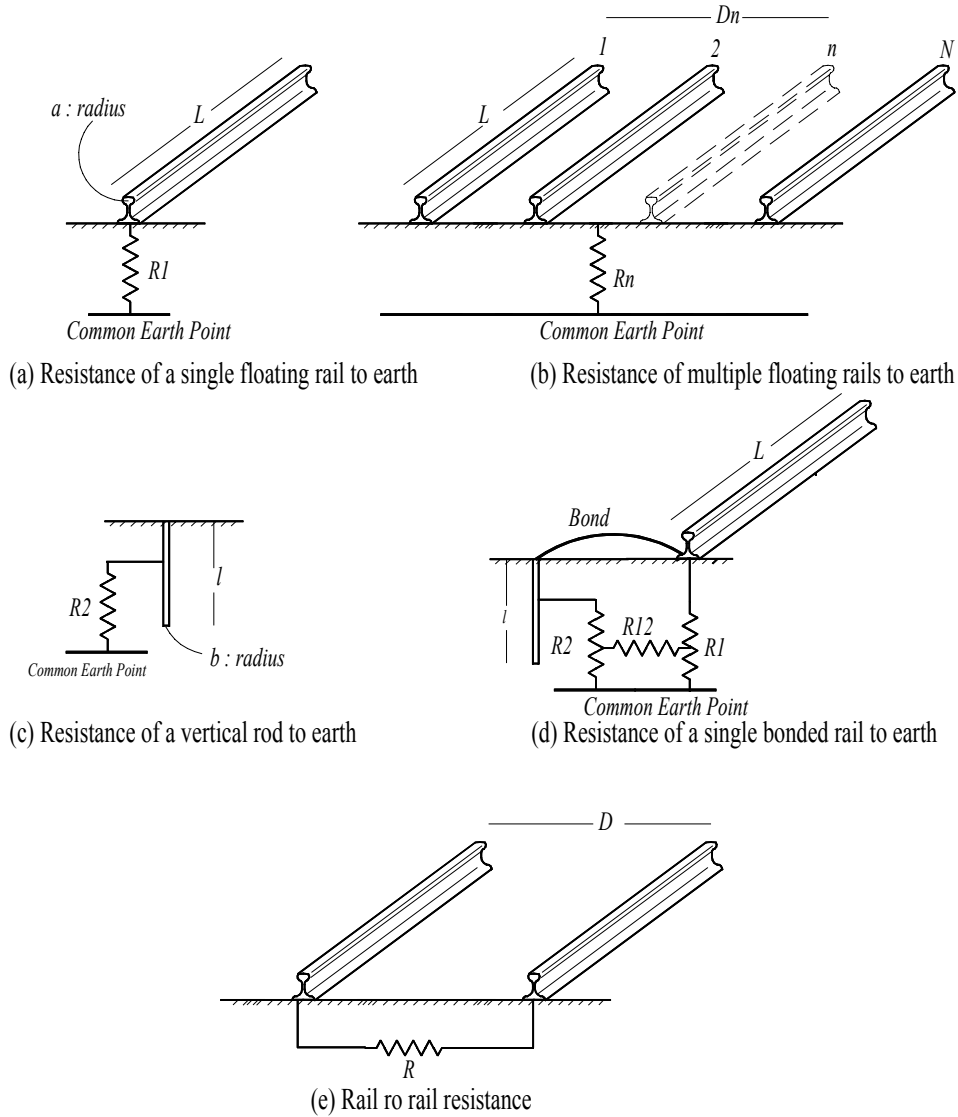


Figure A-3 Different combinations of rail to earth resistance

Resistance of Floating Rails to the Common Earth Point

For a single rail having a length  $L$  and a radius  $a$  the rail to the common earth point resistance, as a function of the rail radius  $a$ , is given by equation 4, where  $\rho$  is the earth resistivity in  $\Omega\cdot\text{m}$ . Refer to Figure A-3(a).

$$R1 = R(a) = \frac{\rho}{\pi L} \left[ \ln \left( \frac{2L}{a} \right) - 1 \right] \quad (4)$$

For  $N$  conductors running in parallel the earth resistance can be derived by expanding the previous formula and approximating it as shown in equation 5. Refer to Figure A-3 (b).

Taking into account the different characteristic values of the earth and the ground than an imaginative plane can be drawn through the axis of the conductor which represents the earth surface. The medium, the conductor and the conductor current to one side of the plane may be removed without disturbing the potential.

$R(a)$  is in fact the equivalent resistance to earth of two rail each of length  $L$  and radius  $a$ , assuming  $a \ll L$ . By adding up all mutual resistances  $R(D_n)$  in case of multiple rails and dividing by the number of the rails the equivalent resistance of the whole system can be found as:

$$R_n = \frac{1}{n} \left[ R(a) + \sum_1^{n-1} R(D_n) \right] \quad (5)$$

where  $D_n$  is the distance between the first conductor and the  $n$ th conductor.

#### Resistance of Earthing and Bonding Points to the Common Earth Point

For an earthing point using a vertical rod of a length  $l$  and a radius  $b$  the earth resistance is defined by equation 6. Refer to Figure A-3 (c).

$$R2 = R(b) = \frac{\rho}{2\pi l} \left[ \ln\left(\frac{4l}{b}\right) - 1 \right] \quad (6)$$

#### Resistance of Earth Bonded Rail to the Common Earth Point

If the rail is earthed at regular intervals by bonding it to the masts which in turn are earthed locally by earthing rods, the rail to earth resistance is calculated using equation 7. Refer to Figure A-3 (d).

$$R = \frac{R1R2 - R12^2}{R1 + R2 - 2R12} \quad (7)$$

where  $R1$  and  $R2$  are given in equation 4 and equation 6 respectively and  $R12$  is the mutual resistance between the rail and the rod given in equation 8

$$R_{12} = \frac{\rho}{\pi L} \left[ \ln \left( \frac{2L}{l} \right) \right] \quad (8)$$

Practically the rail to earth resistance is specified in  $\Omega \cdot \text{km}$  and assumed to be proportional to the earth resistivity  $\rho$ . Furthermore the ratio between the rail to earth resistance for floating rails and earth bonded rails is always constant for the same level of earth resistivity. This is true since the bonding is regularly distributed and the earth resistivity is the same in both cases. There are approximately 16 masts per km on the ac railways. The ratio between the rail to earth resistance for floating rails and earth bonded rails which used in [4] is 1.16.

#### Resistance of the Rail to Rail Earth Surface

The rail to rail resistance is calculated using equation 9 which is derived for buried conductors and takes into account the depth of the burial. Refer to Figure A-3 (e).

$$R = \frac{\rho}{\pi} \ln \left( \frac{D^2}{2ad} \right) \quad (9)$$

where  $D$  is the distance between the two rails and  $d$  is the depth of burial. For a half buried circular conductor it can be shown that the term  $2ad$  can be replaced by  $a$ . This has also been verified against using Laplace's equation where the rail to rail resistance is proportional to the logarithm of the square of the separation distance assuming the earth is a homogenous media.

Calculations have demonstrated that the value ballast of ballast resistivity does not have a significant effect on the calculated current flows in the rails of the neighbouring railway [ref. 7]. A value of 2  $\Omega \cdot \text{km}$  is therefore used.

#### Resistance of Fences to the Common Earth Point

For an earthed fence the fence to the common earth resistance equation 5 can be used.

### **Crossing at an Oblique Angle**

When the two tracks are crossing each other at an oblique angle, either over a bridge or in a tunnel, both the conductive and inductive coupling for the different sections will change. The effect on the conductive coupling is a change in the surface resistance whereby this distance is smallest at the point of intersection (distance here given by the height if the crossing is over a bridge or the altitude difference between the two tracks when crossing is in a tunnel).

At distances further away from the intersection point the distance between the two tracks is governed by the angle of crossing and the distance from the crossing point. The effect of oblique exposures and crossings can be taken into account by using special expressions or values of impedances which are valid for these kind of approaches. The oblique exposures can be subdivided into so-called calculation sections which can be substituted by a series of parallel line sections. In this way the oblique exposure can be calculated from the mutual impedances relating to the parallel lines [6], [15].

### **A3 Shared electrical interface**

The same technique as for a general electrical interface technique can be used to model a shared electrical interface. A coupling factor is determined between the current flow on the Railtrack infrastructure and current flow in the running rails of neighbouring railways. The system model has to be based on the precise bonding arrangements. For example, in case of dc electrification the substation impedance is taken into account. Operating multiple trains at full load results in lower input impedance of the dc substation and consequently lower transfer ratio of the interference currents. Also, different scenarios have to be considered [7] taking into account the input impedance of the trains.

The assessment has to take into account the interference produced during all train operating conditions, including faults.

### **A4 Worked Example**

An example of coupling factors in case of a general electrical interface for both WSF and RSF is presented below as a most frequent case for determining the interaction between neighbouring railways.

The currents induced in the running rails of the neighbouring railways are due to the longitudinal voltages produced by the current flow on the RT infrastructure. Once the longitudinal voltages are found it is simply a matter of calculating the loop current in the circuit of Figure A-2. A case of the two railways running parallel for a distance of 600 m and separation distance of 5 m is considered. The assessment is based on establishing the worst case of train interference and determining the coupling factors between the current flow on the RT infrastructure and current flow in the running rails of the neighbouring railways at the questioned frequency (2kHz in the current example). On RT dc routes the substations are spaced at a maximum distance of 4 km. It is considered that no more than 4 trains would be operating at any time between two adjacent substations. The total train interference current is calculated from the measured levels per unit using the RSS method. The track circuit relay resistance and inductance are accommodated for in the schematic diagram. The rail parameters, including the mutual inductance are calculated as from A2.1 and A2.2. The coupling factors for both WSF and RSF are shown in Table 1:

| Case Study | Train Interference Current at 2 kHz [mA/converter] | Railtrack/LUL Coupling Factor to Rail Current (4 trains) [%] |                  | Neighbouring Authority Limit [mA] | Interference Budget |                  |
|------------|----------------------------------------------------|--------------------------------------------------------------|------------------|-----------------------------------|---------------------|------------------|
|            |                                                    | Normal conditions                                            | Fault Conditions |                                   | Normal conditions   | Fault Conditions |
| WSF        | 24                                                 | 0.97                                                         | 5.2              | 20                                | 25                  | 5.3              |
| RSF        | 24                                                 | 0.97                                                         | 5.2              | 14                                | 21                  | 3.7              |

*Table 1: Coupling Factors and Interference Budget*

The coupling factors calculated for WSF and RSF are the same because they both represent a ratio at rail level rather than at the receiver end. The coupling factor is then used in conjunction with the defined permissible levels by the neighbouring railways to determine the maximum allowed train current on RT infrastructure.

## APPENDIX B: ASSUMPTIONS

**Assumptions:** configuration, supporting references and infrastructure failure modes

### General

- 1 Only conductive and inductive coupling are considered. The effect of capacitive coupling is insignificant and therefore ignored. Resistive coupling can also be ignored if assumption 2 is valid.
- 2 The effect of mutual coupling between the different conductors is calculated using the Carson-Pollaczek method. The proportion of the traction current returning through the earth is assumed to be 30% of the total traction return current to eliminate the imaginary part of the Carson-Pollaczek equation.
- 3 The earth is assumed to be a homogenous media of a constant resistivity,  $\rho$ , given in  $\Omega\cdot\text{m}$ . The ballast is assumed to have the same resistivity. The indicative value used in the calculations is  $300\ \Omega\cdot\text{m}$ , [6] and [7] and it is a representative figure for large area of the country [18].
- 4 Ballast resistance does not make a significant difference to the current interference levels in the running rails of neighbouring railways [7]. The value of  $2\ \Omega\cdot\text{km}$  is used in the example modelling exercise.
- 5 Rail to earth resistance affects significantly the interference levels [7]. A factor of 0.8 gives a good approximation to the equivalent rail to earth resistance of two running rails.
- 6 Two earth leakage paths between the running rails are considered; one is between the rails and a reference earth point and the other is between the nearest two rails of the two railways.



## **APPENDIX C: SOURCES OF TECHNICAL WORK**

The technical work contained in this document is taken from a number of sources:

- The Class 92 EMC safety case, produced by AEA Technology Rail, ISAs were WS Atkins Rail (secretariat reviewer) and Mott MacDonald.

The Class 375 "Electrostar" EMC safety case, produced by Bombardier, ISA was AEA Technology Rail.

# RAILTRACK

---

## BRIEFING NOTE

**RT/E/C/50001 to RT/E/C/50019**

**Issue 1**

**Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure**

**Issue Date: February 2003**

---

### **Background / Synopsis**

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

### **Key Changes / Compliance**

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

### **Implementation**

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

### **Of Interest To:**

Authors of Route Acceptance Safety Cases.

### **Name and Title of Author**

David Wilkinson  
Technical & Safety Advisor